

DEPARTMENT OF CIVIL ENGINEERING

Indian Institute of Technology (Indian School of Mines), Dhanbad



A MINI PROJECT REPORT ON TRAFFIC VOLUME SURVEY & ROAD SAFETY ANALYSIS

A Case Study of IIT (ISM) Main Gate

Submitted in partial fulfilment of the requirements for
Bachelor of Technology (B.Tech.) in Civil Engineering

CIVIL ENGINEERING

By

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CERTIFICATE

This is to certify that the project report entitled "Traffic Volume Survey & Road Safety Analysis – A Case Study of IIT (ISM) Main Gate" is a bona fide record of work carried out by Brajbhushan Sharma, Roll No. 24JE0016, in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology (B.Tech.) in Civil Engineering.

The work embodied in this report has been carried out under my supervision and has not been submitted elsewhere for the award of any other degree or diploma.

Date: 24/05/2026

Place: Dhanbad

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Brajbhushan Sharma

ABSTRACT

Traffic congestion and road safety are among the most pressing challenges faced by rapidly growing urban and semi-urban areas in India. This project presents a field-based Traffic Volume Survey and Road Safety Analysis carried out at IIT (ISM) Main Gate, with the objective of understanding traffic flow characteristics, identifying safety deficiencies, and proposing practical engineering interventions.

A manual traffic volume count was conducted during morning (8:00–9:00 AM) and evening (5:00–6:00 PM) peak hours on two separate days, classifying vehicles into two-wheelers, cars, auto-rickshaws, buses, and trucks at 15-minute intervals. The collected data was analysed in Microsoft Excel to determine hourly traffic volume, vehicle composition, peak-hour flow, and the level of congestion.

In parallel, a road safety assessment was performed covering pedestrian crossings, traffic signage, street lighting, footpaths, parking discipline, and helmet/seat-belt compliance, supported by photographic documentation. The study found that two-wheelers dominate the traffic composition, accounting for close to 59% of total volume, and that inadequate pedestrian infrastructure, illegal parking, and poor lane discipline are the primary contributors to congestion and safety risk at the location.

Based on these findings, the report recommends short-term and long-term engineering measures including provision of a marked pedestrian crossing, improved lane markings, better street lighting, removal of encroachments, and enhanced enforcement during peak hours.

Keywords: Traffic Volume Survey, Road Safety Assessment, Peak Hour Analysis, Vehicle Classification, Traffic Engineering, Transportation Planning.

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INTRODUCTION

Background

Rapid urbanisation and rising vehicle ownership have led to increasing traffic volumes on urban and semi-urban roads across India, often outpacing the capacity of existing infrastructure. This mismatch results in recurring congestion, delays, and a heightened risk of road accidents, particularly at intersections and mixed-traffic corridors where two-wheelers, cars, auto-rickshaws, buses, and trucks share the same carriageway.

The IIT (ISM) Main Gate experiences a heavy mix of institutional, residential, and commercial traffic, especially during morning and evening peak hours. Recurring congestion, informal parking, and gaps in pedestrian infrastructure motivated a focused field investigation to quantify traffic characteristics and document existing safety deficiencies at this location.

Need for the Study

Traffic volume studies form the foundation of transportation planning and traffic engineering. By quantifying how many vehicles of each type use a road section during specific time periods, engineers can identify peak demand, assess capacity adequacy, and design appropriate interventions such as improved signal timing, lane discipline measures, or pedestrian facilities.

Scope of the Study

The scope of this study is limited to a single road location, surveyed manually over two days during defined peak periods. The study does not include automated traffic counting equipment or vehicle speed measurement instruments, and is intended as an academic field exercise demonstrating core traffic engineering and data analysis skills.

OBJECTIVES

The objectives of this study are as follows:

- To conduct a manual traffic volume survey and classify vehicles by type during peak hours.
- To determine hourly traffic volume, vehicle composition, and the busiest time interval.
- To assess the level of congestion at the study location.
- To carry out a road safety assessment covering pedestrian facilities, signage, lighting, and enforcement-related observations.
- To identify key road safety problems and propose practical, low-cost engineering recommendations.

STUDY AREA

Location Description

The study was conducted at IIT (ISM) Main Gate, a mixed-traffic location characterised by institutional, residential, and commercial land use in the surrounding area. The location was selected because it experiences a representative mix of vehicle types and observable congestion during peak hours.

Site Characteristics

- Carriageway width: 11.2 m
- Number of lanes: 2-lane, two-way
- Adjoining land use: Mixed land use (institutional, residential, and commercial)
- Nearby landmarks: Refer to the Location Map (Fig. 1 and Fig. 2) for landmarks visible in the surrounding area.
- Type of control: Regulated by traffic police during peak hours, with signal control

LOCATION MAP

The following figures show the geographic location of the study area, obtained from Google Maps.

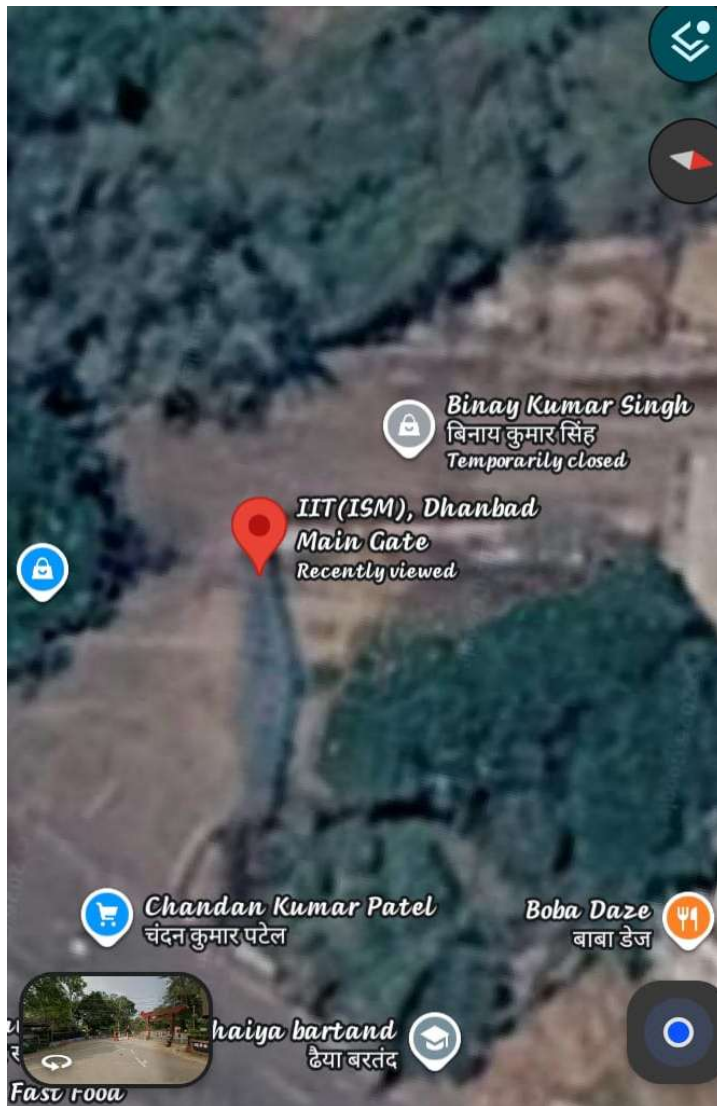


Figure 1: Google Maps – Study Area (Wide View)

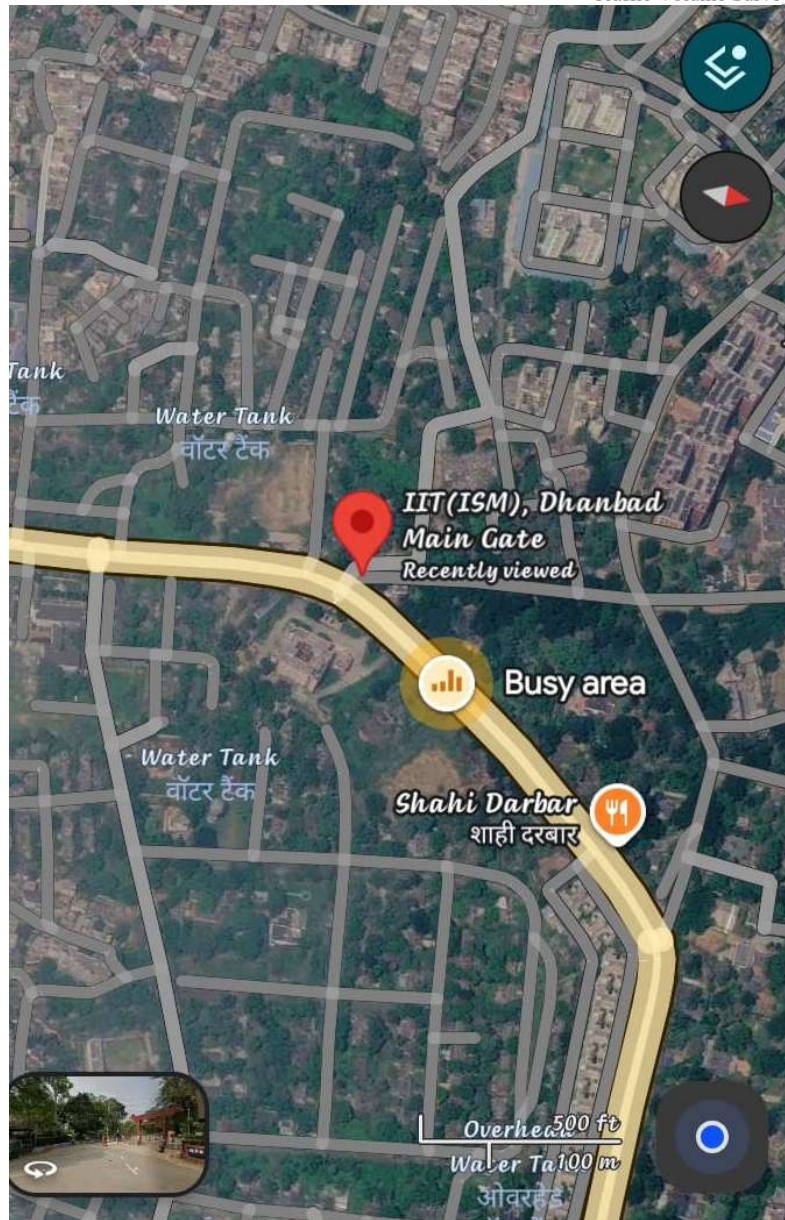


Figure 2: Google Maps – Detailed Study Area

METHODOLOGY

Survey Design

A manual classified traffic volume count was adopted, given its simplicity, low cost, and suitability for a short-duration academic field study. Two observers were stationed at the survey point with tally sheets, a stopwatch, and a camera.

Survey Schedule

- Morning peak session: 8:00 AM – 9:00 AM
- Evening peak session: 5:00 PM – 6:00 PM
- Survey conducted on two separate days to improve reliability of results
- Counts recorded manually at 15-minute intervals

Vehicle Classification

Vehicles were classified into the following categories for ease of counting and analysis:

- Two-Wheelers (motorcycles, scooters)
- Cars (including SUVs and vans)
- Auto-Rickshaws
- Buses
- Trucks (including LCVs and HCVs)
- Others (bicycles, tractors, animal-drawn carts, etc.)

Field Equipment

- Tally sheets / count boards (prepared in advance)
- Stopwatch / mobile timer for 15-minute intervals
- Camera / smartphone for photographic documentation
- Measuring tape (for carriageway width)

Road Safety Assessment Procedure

Alongside the volume count, a walk-through road safety assessment was performed to visually assess the presence and condition of pedestrian crossings, traffic signals, speed breakers, street lighting, footpaths, and parking discipline, as well as compliance with helmet and seat-belt usage. Observations were recorded on a checklist and supported with photographs.

Data Analysis Tools

- Microsoft Excel — tabulation, percentage computation, and chart generation
- Google Maps — study area location and layout reference
- AutoCAD (optional) — schematic site layout, if required

Existing Site Conditions

The IIT (ISM) Main Gate is served by a 2-lane, two-way carriageway of approximately 11.2 m in width. The surrounding area has mixed land use (institutional, residential, and commercial). Traffic movement at the location is regulated by traffic police during peak hours, with signal control.

Observed Infrastructure Elements

- Carriageway: paved, two-lane two-way, with visible wear in sections.
- Signage: a mix of regulatory and directional signs, some faded or missing.
- Street lighting: functional along the main stretch, sparse near the junction.
- Drainage: covered drains along most of the stretch, with isolated open sections.
- Speed breakers: present near the main gate, partially worn.

Figure 3: Main Entrance / Gate — IIT (ISM) Main Gate



Figure 4: Overall View of Study Area



Figure 5: Existing Road Geometry



Figure 6: Traffic / Regulatory Sign Board



Figure 7: Zebra Crossing



Figure 8: Street Lighting Arrangement

FIELD OBSERVATIONS

Morning Peak Hour Data (Day 1, 8:00 AM – 9:00 AM)

Table 1 presents the classified vehicle count recorded at 15-minute intervals during the morning peak session.

Time Interval	Two-Wheeler	Car	Auto	Bus	Truck	Total
8:00 – 8:15 AM	96	34	19	3	11	163
8:15 – 8:30 AM	101	41	22	2	13	179
8:30 – 8:45 AM	112	38	20	4	10	184
8:45 – 9:00 AM	107	36	18	3	12	176
Total	416	149	79	12	46	702

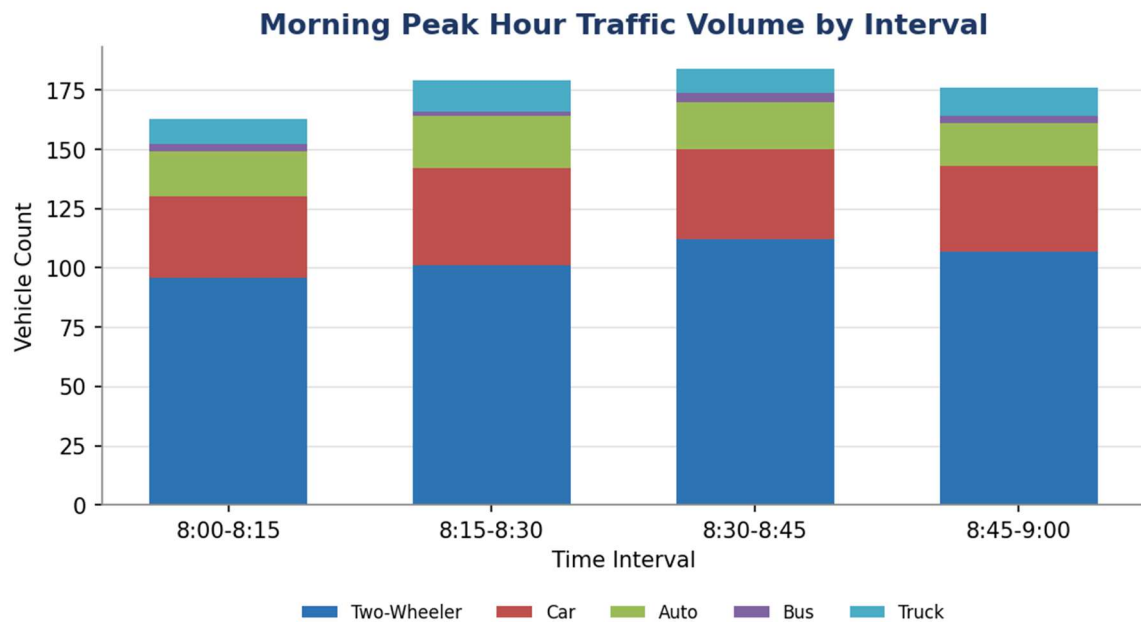


Chart 1: Morning Peak Hour Traffic Volume by Interval

Evening Peak Hour Data (Day 2, 5:00 PM – 6:00 PM)

Table 2 presents the classified vehicle count recorded during the evening peak session on the second survey day.

Time Interval	Two-Wheeler	Car	Auto	Bus	Truck	Total
5:00 – 5:15 PM	89	31	17	4	9	150
5:15 – 5:30 PM	98	35	21	3	12	169
5:30 – 5:45 PM	121	40	24	5	13	203
5:45 – 6:00 PM	104	37	19	2	11	173

Time Interval	Two-Wheeler	Car	Auto	Bus	Truck	Total
Total	412	143	81	14	45	695

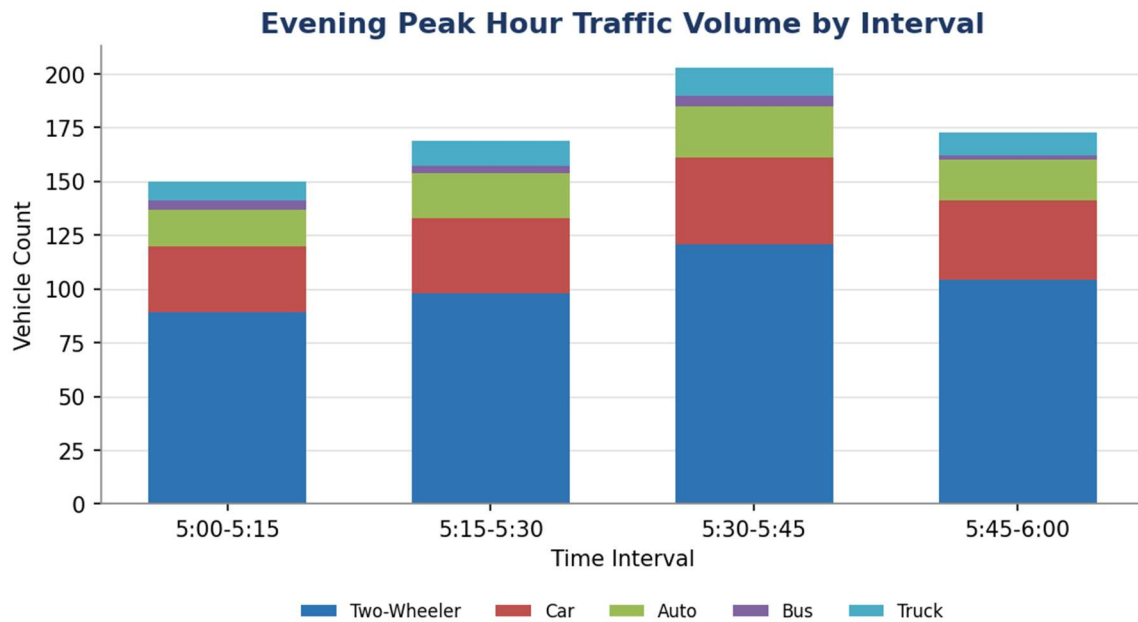


Chart 2: Evening Peak Hour Traffic Volume by Interval

Field Photographs

The following photographs document the prevailing traffic conditions during the survey.



Figure 9: Traffic Movement During Morning Peak Hour



Figure 10: Traffic Movement During Evening Peak Hour

TRAFFIC ANALYSIS

Total Traffic Volume

The total traffic volume recorded during the morning peak hour (8:00–9:00 AM) was 702 vehicles/hour. The total traffic volume during the evening peak hour (5:00–6:00 PM) was 695 vehicles/hour, indicating comparable demand during both peak periods.

Vehicle Composition (Morning Peak)

Table 3 shows the percentage share of each vehicle type in the total morning peak-hour traffic.

Vehicle Type	Count (veh/hr)	Percentage
Two-Wheelers	416	59.3%
Cars	149	21.2%
Auto-Rickshaws	79	11.3%
Buses	12	1.7%
Trucks	46	6.6%

Two-wheelers form the dominant share of traffic at approximately 59.3%, followed by cars at 21.2% and auto-rickshaws at 11.3%. Buses and trucks together account for less than 9% of total volume, indicating a predominantly personal and intermediate-transport oriented traffic mix typical of institutional/residential corridors.

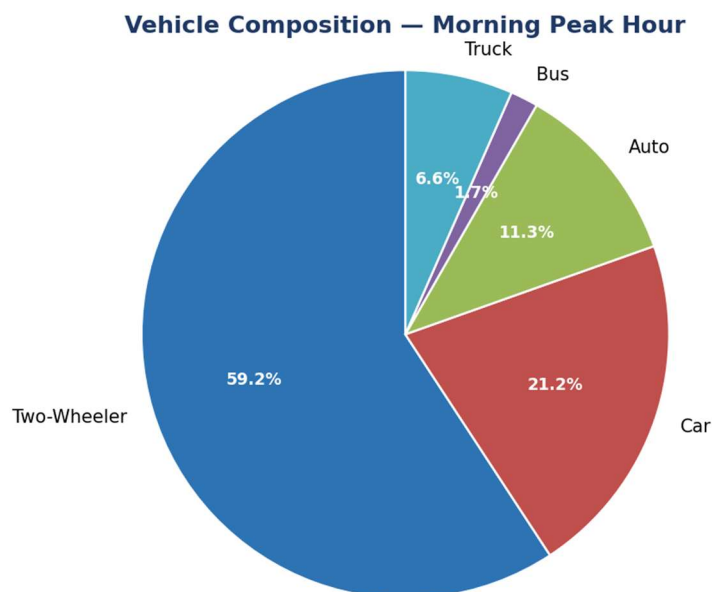


Chart 3: Vehicle Composition — Morning Peak Hour

Peak Hour Factor (PHF)

Peak Hour Factor (PHF) is a traffic engineering parameter used to measure the variation of traffic flow during the peak hour. It indicates whether the traffic volume is evenly distributed throughout the hour or concentrated within a short time interval. A PHF value closer to 1.0 represents a more uniform traffic flow, whereas a lower value indicates significant fluctuations during the peak period.

Formula

$$\text{Peak Hour Factor (PHF)} = \frac{\text{Hourly Traffic Volume}}{4 \times \text{peak 15-minute traffic volume}}$$

Calculation

For the morning peak hour survey,

- Hourly Traffic Volume = 702 vehicles/hour
- Peak 15-Minute Traffic Volume = 184 vehicles

$$\text{PHF} = \frac{720}{4 \times 184} = 0.954$$

Interpretation

The calculated Peak Hour Factor (PHF) is 0.954, which indicates that the traffic flow during the morning peak hour was fairly uniform, with only moderate variation between successive 15-minute intervals. This suggests that traffic demand remained consistently high throughout the survey period rather than being concentrated within a single short-duration peak.

Peak 15-Minute Interval

The busiest 15-minute interval was observed between 8:30 AM and 8:45 AM, with a recorded volume of 184 vehicles, corresponding to an hourly equivalent of approximately 736 vehicles/hour (184×4), which represents the highest short-duration demand within the survey period.

Congestion Level Assessment

Based on the observed volume-to-capacity conditions, the queuing behaviour noted at the location, and the width of the carriageway, the congestion level during peak hours is assessed as follows:

- Morning Peak (8:30–8:45 AM): High congestion
- General Morning Session: Moderate to High congestion
- Evening Peak (5:30–5:45 PM): High congestion
- Off-peak periods (not surveyed): Expected Low congestion

Discussion

The dominance of two-wheelers combined with the absence of dedicated lane discipline contributes significantly to the observed congestion, as two-wheelers frequently weave between larger vehicles, reducing effective lane capacity. The presence of on-street parking and roadside encroachment further narrows the usable carriageway width, compounding delays during peak periods.

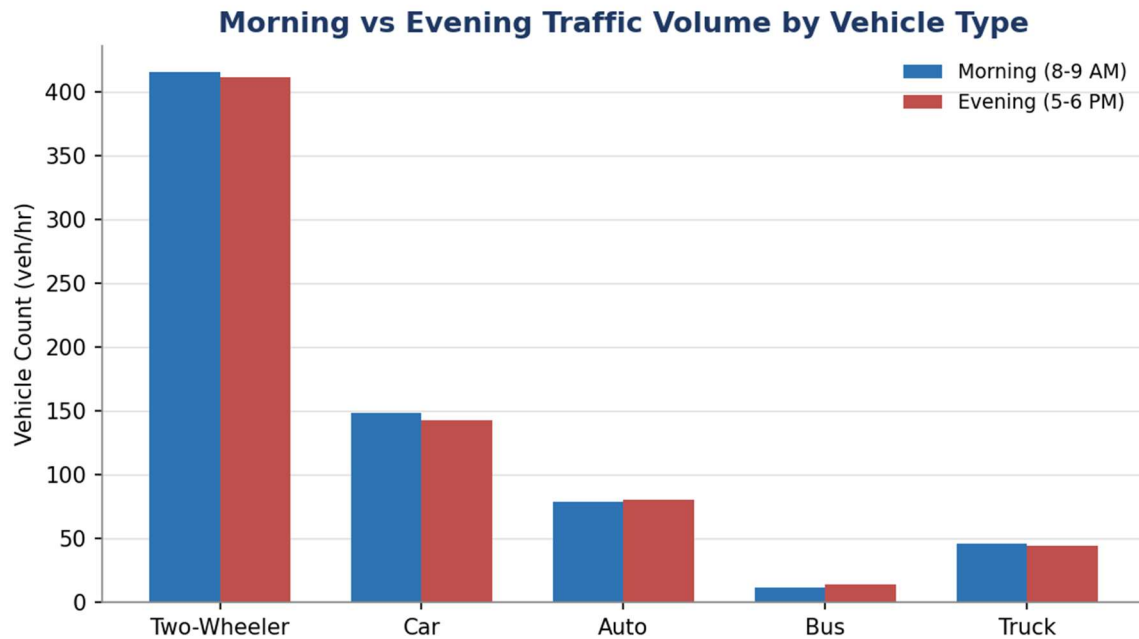


Chart 4: Morning vs Evening Traffic Volume by Vehicle Type

Road Safety Audit

Assessment Checklist and Observations

Table 4 summarises the road safety features observed at the study location during the field visit.

Safety Feature	Observed Status	Remarks
Zebra Crossing	faded	Present near main gate
Traffic Signal	Present, active during peak hours	Timing review recommended
Speed Breakers	Present, partially worn	Repainting required for visibility
Street Lighting	Functional, sparse near junction	Additional poles recommended
Footpath	Discontinuous / encroached	Needs continuous, obstruction-free path
Illegal Parking	Observed on both sides	Enforcement and designated bays needed

Safety Feature	Observed Status	Remarks
Helmet Usage	Moderate compliance (~65%)	Awareness drive recommended
Seat Belt Usage	High compliance (~85%)	Satisfactory
Drain Covers	Mostly closed, 1–2 open	Immediate repair needed
Potholes	Minor potholes observed	Patch repair recommended
Encroachment	Roadside vendors present	Affects effective carriageway width

Key Road Safety Problems Identified

- Absence of a clearly marked pedestrian crossing near the main access point.
- Poor lane discipline, particularly among two-wheeler riders during peak hours.
- Frequent over speeding by two-wheelers approaching the location.
- Illegal roadside parking reducing the effective carriageway width.
- Street vendors and informal encroachment further narrowing usable road space.
- Missing or faded road markings, reducing lane guidance for drivers.
- Inadequate street lighting in sections away from the main junction.

Photographic Record of Safety Observations

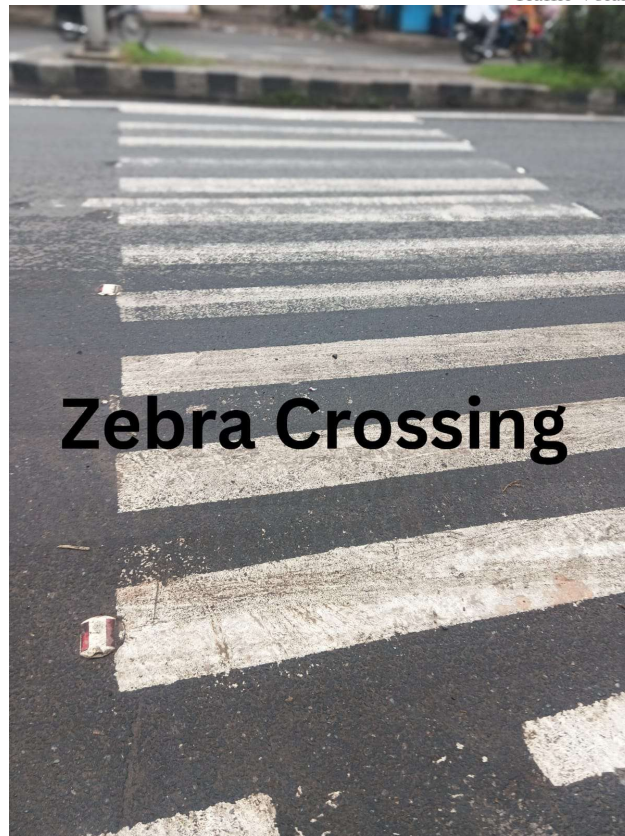


Figure 11: Zebra Crossing / Pedestrian Crossing Point



Figure 12: Footpath Condition



Figure 13: Roadside / Illegal Parking

ENGINEERING RECOMMENDATIONS

Short-Term (Low-Cost) Measures

- Install a clearly marked zebra crossing at the main pedestrian conflict point.
- Repaint faded lane markings and provide edge lines to improve lane discipline.
- Install convex mirrors at sight-restricted points to improve visibility.
- Install warning and regulatory signage (speed limit, no-parking zones).
- Increase traffic police presence during identified peak hours (8:30–8:45 AM and 5:30–5:45 PM).

Medium-Term Measures

- Improve street lighting along sections identified as inadequately lit.
- Provide a continuous, obstruction-free footpath for pedestrians.
- Designate formal parking bays to discourage illegal roadside parking.
- Relocate or regulate street vendors encroaching on the carriageway.

Long-Term Measures

- Review and optimise signal timing as traffic volumes continue to grow.
- Consider channelisation or minor geometric redesign to separate conflicting movements.
- Periodic (annual) traffic volume monitoring to track growth trends and revisit interventions.

SWOT ANALYSIS

A SWOT analysis was carried out to summarise the internal and external factors affecting traffic operations and safety at the study location.

Strengths

- Well-connected location with a good mix of vehicle types, useful for representative data collection.
- Reasonably wide carriageway that can accommodate engineering improvements without major reconstruction.
- High seat-belt compliance among car users (~85%), indicating good baseline safety awareness.

Weaknesses

- Absence of a marked pedestrian crossing.
- Discontinuous footpath forcing pedestrians onto the carriageway.
- Encroachment by informal vendors and illegal parking.

Opportunities

- Low-cost interventions (markings, signage, crossings) can meaningfully improve safety.
- Potential for coordination with local traffic police for peak-hour regulation.
- Scope for pedestrian infrastructure improvement given available right-of-way.

Threats

- Continued growth in two-wheeler ownership may worsen congestion without intervention.
- Unregulated encroachment could expand further, reducing effective carriageway width over time.
- Delay in implementing safety measures increases accident risk, particularly for pedestrians.

CONCLUSION

This study successfully carried out a classified traffic volume survey and road safety assessment at IIT (ISM) Main Gate, applying standard traffic engineering methodology within a short field-study duration. The results show that the location experiences a peak traffic volume of approximately 700 vehicles/hour, dominated by two-wheelers (~59%), with the busiest interval occurring between 8:30 and 8:45 AM.

The road safety assessment revealed several deficiencies, most notably the absence of a marked pedestrian crossing, discontinuous footpaths, illegal parking, and inadequate street lighting in certain sections. These factors collectively increase both congestion and accident risk at the location.

Based on the findings, a set of prioritised, low-cost engineering recommendations has been proposed, including installation of a pedestrian crossing, improved lane markings, better signage, and enhanced peak-hour enforcement. Implementation of these measures is expected to improve both traffic flow efficiency and pedestrian safety at the study location.

This project demonstrates the practical application of traffic engineering and transportation planning concepts, and reflects hands-on skills in field survey design, data collection, data analysis using MS Excel, and technical report writing.